

ANDROID DEVELOPER INTERNSHIP PROGRAM

Task 5

Task Title:

Create Database Schema & ER Diagram (Room / Firebase)

App Name: Eco-Cart

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1. Identify Entities (Collections)

For Eco-Cart, we need three main collections to handle users, their history, and the product data.

1. **Users:** Stores profile info and total environmental impact.
2. **Products:** The master database of items and their eco-scores.
3. **Scans:** A "linking" collection that records every time a specific user scans a specific product.

2. Define Attributes (Fields)

In Firebase, every document gets a unique ID. For Products, we will use the **Barcode** as the Document ID.

Collection	Field Name	Data Type	Description
Users	uid	String (Key)	Unique ID from Firebase Auth
	name	String	User's full name
	email	String	User's email address
	totalCO2Saved	Double	Cumulative environmental points
Products	barcode	String (Key)	The scanned UPC/EAN code
	name	String	Product Name (e.g., "Organic Milk")
	ecoScore	String	Rating (A, B, C, D, or E)
	carbonFootprint	Double	Numerical value of CO2 impact

	category	String	e.g., "Beverages"
Scans	scanId	String (Key)	Unique ID for this event
	userId	String (FK)	Reference to the User who scanned
	productId	String (FK)	Reference to the Barcode scanned
	timestamp	Timestamp	Date and time of the scan

3. Define Relationships

Even in NoSQL, we must define how data relates:

- **One-to-Many (User → Scans):** One user can scan many products over time.
- **Many-to-One (Scans → Product):** Many different scan events can point to the same product in the database.
- **One-to-One (User → Auth):** Each user document in Firestore corresponds to one account in Firebase Authentication.

4. Database Schema (Firebase Structure)

1. **Root Collection: /users/{uid}** (Each user has their own document containing their profile and stats.)
2. **Root Collection: /products/{barcode}** (Global list of products. Using the barcode as the ID makes searching extremely fast.)
3. **Root Collection: /scans/{scanId}** (Flat collection for history. We use userId as a filter to show a specific user's history.)

5. ER Diagram Reference (Draw.io)

